MCMC-Based Road Profile Estimation for Adaptive Suspension Mode Selection

Estimation & Filtering Homework Project

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1 Introduction: Project Motivation and Idea

Modern road vehicles, especially sedans and coupes leaning toward the executive and luxury class, provide several driving mode configurations (which physically amount to varying the vehicle height and suspension stiffness) to enable a more ergonomic experience for the driver and passengers on board. Manually changing the driving settings on the go can be unintuitive and impractical, and the implementation of a switching mechanism between driving modes cannot rely on navigation data alone, as this data is often not up-to-date with real terrain conditions.

The second reason makes the design of an Estimator that uses the vehicle's surrounding sensor data to build a real-time road profile, which helps to adaptively select the driving mode, a natural candidate for providing the vehicle state to the car's Electronic Control Unit (ECU).

This work is in line with the concepts studied in the E&F course and has grounds for real industry-level application.

Figure 1: Example of Audi manufacturer's *Drive Select* infotainment interface for selecting between - and customizing - driving modes. Source: Audi New York

The proposed goal of this homework project is to use Markov Chain Monte Carlo (MCMC) to estimate the roughness parameters of a road profile from noisy vehicle measurements, which is then used to select a suitable driving/suspension mode.

2 Road Profile: Data Generation and Modeling

The generation mechanism for the road profiles used for the simulations needed to provide enough variety and enable the lack of reproducibility of the exact same road profile between any two trials[1], so that the MCMC Estimator and the overall Controller (ECU) of the car are tested in robust, realistic conditions. To this end, we chose to implement the model proposed by Rill in [1]. This model endows a Random Stationary Gaussian Process to the vertical deviation of the road surface:

$$z_R = z_R(s), \quad \text{with } s \text{ travelled distance along the road}$$

Therefore, $z_R(s)$ represents the deviation of the road surface from a *zero* (smooth) reference surface. Positive values of $z_R(s)$ correspond to upward irregularities, such as bumps, while negative values correspond to downward irregularities, such as potholes or depressions.

[1] As can be seen in the simulation files, if reproducing the same results is actually needed, a *seed* to the random processes is provided with the command `rng(x)`; where `x` holds the *key* to a particular realization.

2.1 Road Profile as a Random Stationary Gaussian Process

Under this formulation, the assumption of Gaussianity means that, at a fixed position s, the random variable of road height can be described by a normal distribution:

$$z_R(s) \sim \mathcal{N}(m, \sigma^2), \quad \text{with } m \text{ mean value and } \sigma^2 \text{ variance of road height.}$$

The mean road elevation can be assumed to be zero. This does not mean that the road is flat, but only that the road profile fluctuates around a reference level, more conveniently chosen as zero. The variance is then an indicator of the roughness of the road profile, where larger variance corresponds to a rougher road profile, while a smaller variance corresponds to a smoother road profile.

2.2 Autocorrelation Function

The Gaussian distribution describes the statistical distribution of the road elevation values, but it does not describe how the road varies along the distance coordinate s. For that, the autocorrelation function is introduced:

$$R(\xi) = \mathbb{E}\{z_R(s)z_R(s+\xi)\}, \quad \text{with } \xi \text{ spatial shift following the direction of the road.}$$

The autocorrelation function compares the road height at one point, $z_R(s)$, with the road height at another point, $z_R(s+\xi)$, located ξ meters away. For $\xi = 0$, the autocorrelation becomes $R(0) = \mathbb{E}\{z_R^2(s)\} = \sigma^2$.

Therefore, the autocorrelation at zero shift is equal to the variance of the road profile. For larger values of ξ, the connection between $z_R(s)$ and $z_R(s+\xi)$ usually becomes weaker. In a typical random road profile, distant points are less related than nearby points, meaning that $R(\xi)$ will tend to 0 as ξ tends to infinity.

The autocorrelation function can also reveal periodic components of road height if they are present inside the random profile, such as regularly spaced bumps, paving joints, or wave-like undulations.

2.3 Power Spectral Density

Road profiles are usually characterized in the spatial frequency domain using the power spectral density, or PSD. The PSD is related to the autocorrelation function through the Fourier transform:

$$S(\Omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} R(\xi)e^{-i\Omega\xi}d\xi, \quad \text{and its inverse relation } R(\xi) = \int_{-\infty}^{+\infty} S(\Omega)e^{i\Omega\xi}d\Omega.$$

Here Ω is the spatial wave number, measured in $[\mathrm{rad/m}]$. Just like temporal frequency describes oscillations per second in the time domain, in road profiles, characterized in the spatial domain, wave number describes oscillations per meter. The relation between wave number and wavelength is:

$$\Omega = \frac{2\pi}{L}, \qquad \text{with } L \text{ wavelength in meters.}$$

Therefore a small Ω corresponds to a large L, meaning long and smooth road undulations, and on the contrary, a large Ω corresponds to a small L, meaning short and rapid road irregularities.

The PSD tells how much road roughness variance is present at each spatial wave number. Therefore, a high PSD value at low Ω means that the road contains strong long-distance elevation variations. A high PSD value at high Ω would mean that the road contains strong short-distance roughness, such as sharp bumps or closely spaced irregularities. And equivalently, at either high or low values of Ω, low PSD values would indicate a weak presence, or content, of those spatial frequency ranges. The total variance of the road profile can be obtained by integrating the PSD:

$$\sigma^2 = \int_0^\infty \Phi(\Omega) d\Omega,$$

where $\Phi(\Omega)$ is the one-sided PSD, obtained from the side-stacked PSD $S(\Omega)$ by collecting the positive and negative wave-number contributions into the nonnegative wave-number range:

$$\begin{cases} \Phi(\Omega) = 2S(\Omega) & \text{if} \quad \Omega \geq 0 \\ \Phi(\Omega) = 0 & \text{otherwise.} \end{cases}$$

The equation of the total roughness variance of the road is obtained by infinitestimally summing the contributions of all spatial wave numbers. As will be seen later, also in our implementation, this empirically amounts to a finite sum.

2.4 International Standard Road Roughness Modeling and Classification

In this project, the road PSD is modeled using the ISO 8608 power-law approximation

$$\Phi(\Omega) = \Phi_0 \left(\frac{\Omega}{\Omega_0} \right)^{-w}$$

The PSD therefore has a power-law decay for $w > 0$, and has several parameters that allow its configuration:

- Ω: spatial wave number, in $\mathrm{rad/m}$;
- Ω_0: *reference wave number*, usually chosen as $\Omega_0 = 1\ \mathrm{rad/m}$;
- $\Phi_0 = \Phi(\Omega_0)$: *reference PSD value* at Ω_0;
- w: *waviness parameter*, controlling the slope of the PSD curve.

The parameter Φ_0 controls the overall roughness level of the road and maps us to the road class to be created. Larger values of Φ_0 shift the PSD curve upward and correspond to rougher roads. Smaller values of Φ_0 shift the PSD curve downward and correspond to smoother roads.

The parameter w controls how quickly the PSD decreases as Ω increases. For a fixed Φ_0, so for a chosen road class, larger w suppresses high-Ω components more strongly, while smaller w allows for more high-Ω content. In practice, w acts as a tuning knob to the roll-off of the bandwidth of $\Phi(\Omega)$. Taking the logarithm of the PSD model gives

$$\log \Phi(\Omega) = \log \Phi_0 - w \log \left(\frac{\Omega}{\Omega_0} \right).$$

Thus, on a log-log plot, w determines the slope, while Φ_0 determines the vertical offset. The value Φ_0 is not an Ω-intercept, but rather the PSD value at the reference wave number Ω_0. For the common case $w = 2$, the road classes have the same slope and are distinguished mainly by different values of Φ_0. A qualitative classification is shown in Table 1 and these categories are graphically distinguished on a PSD plot in Figure 2.

Table 1: Qualitative interpretation of ISO-style road roughness classes.

Road Class	Φ_0 [m^2/(rad/m)]	Qualitative Meaning	Typical Interpretation
A	1×10^{-6}	Very smooth road	High-quality highway, smooth asphalt
B	4×10^{-6}	Smooth road	Well-maintained, moderate road
C	16×10^{-6}	Average road	Normal urban or regional road
D	64×10^{-6}	Rough road	Damaged or poorly maintained road
E	256×10^{-6}	Very rough road	Severely damaged or unpaved road

Following this standard, in the project the unknown road parameter vector is chosen as

$$\boldsymbol{\theta} = \begin{bmatrix} \Phi_0 \\ w \end{bmatrix},$$

as it provides a compact and effective way of characterizing a road profile.

The estimator is required to infer the roughness magnitude and the spectral slope of the road profile, thus providing a sufficiently complete representation of a road that can be used reliably to determine the appropriate drive settings by the driving mode selector.

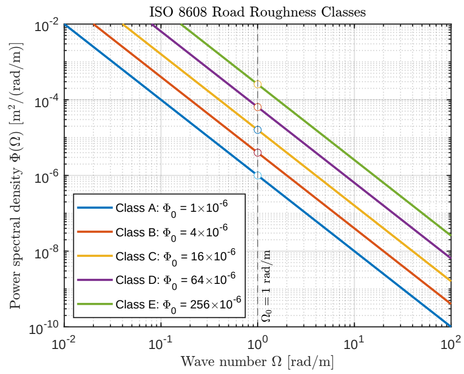


Figure 2: ISO 8608 road roughness classification PSD Plot. The classes share the same waviness slope $w = 2$, while different values of $\Phi_0 = \Phi(\Omega_0)$, evaluated at $\Omega_0 = 1$ rad/m, shift the PSD curves vertically and represent increasing road roughness from Class A to Class E.

2.5 Sinusoidal Approximation of the Road Profile

To generate a road profile in the simulation, the continuous PSD is approximated by a finite sum of sinusoidal components:

$$z_R(s) = \sum_{i=1}^{N} A_i \sin(\Omega_i s - \psi_i)$$

Where Ω_i is the wave number of the i-th component,A_i the amplitude of the i-th component and ψ_i a random phase angle sampled form a uniform distribution as $\psi_i \sim \mathcal{U}(0, 2\pi)$.

The wave numbers are chosen over a finite interval $\Omega_i \in [\Omega_{\min}, \Omega_{\max}]$ that is discretized into N wave numbers with a uniform spacing, so that

$$\Delta\Omega = \frac{\Omega_{\max} - \Omega_{\min}}{N - 1}$$

represents the width of the small wave-number band associated with each sinusoidal component. For each wave number Ω_i, the PSD value is computed using the ISO-8608 model, and the amplitude A_i selected so that the sinusoidal approximation has the desired PSD. The variance contribution of the i-th PSD interval is approximately $\Phi(\Omega_i)\Delta\Omega$. On the other hand, the average squared value of a sinusoid of amplitude A_i is $A_i^2/2$. To make the sinusoidal component match the variance contribution of the PSD interval, we impose:

$$A_i = \sqrt{2\Phi(\Omega_i)\Delta\Omega}$$

Therefore, the empirical sinusoidal road realization is:

$$z_R(s) = \sum_{i=1}^{N} \sqrt{2\Phi(\Omega_i)\Delta\Omega}\,\sin(\Omega_i s - \psi_i).$$

This construction gives a random-looking road profile whose statistical roughness properties are controlled by the PSD model.

Adding Potholes: In the rough-road scenario, localized potholes are additionally superimposed on the stochastic profile as deterministic, negative, cosine-shaped disturbances. These events are not part of the stationary Gaussian PSD model, but contribute to more realistic road profiles (especially for classes D and E) since potholes are sudden and large abruptions from a smooth profile, concentrated in a small spatial interval, which is very rarely generated by the PSD model alone.

2.6 Shaping Filter for Road Profile

Another way of generating a Road Profile is using a first-order stochastic shaping filter driven by white noise. The continuous spatial-domain model is:

$$\frac{d}{ds} z_R(s) = -\Gamma z_R(s) + w(s)$$

where Γ controls the spatial correlation of the profile, and $w(s)$ is a white Gaussian noise process with spectral density Φ_w. This model produces a road whose PSD is low-pass filtered, therefore replicating the dynamic behavior seen in the original implementation. The model is then discretized using the Euler-Maruyama method. Defining the *spatial step* as:

$$\Delta s_k = v_k dt$$

the discrete update becomes

$$z_R[k+1] = z_R[k] - \Gamma z_R[k] \Delta s_k + \sqrt{\Phi_w \Delta s_k} \eta_k$$

where $\eta_k \sim \mathcal{N}(0, 1)$. The stochastic term injects spatially white roughness, while the deterministic term smooths the profile according to Γ.

2.7 Road Generation Procedure

The road generation procedure is implemented in the simulation by initially hand-selecting a total drive time, and a sequence of smooth, average and rough asphalt road segments to be travelled through in this drive time duration. For each different road segment, the road roughness parameters, the wave-number interval, and the N number of discretization intervals are assigned:

$$\boldsymbol{\theta} = \begin{bmatrix} \Phi_0 \\ w \end{bmatrix} \qquad \left[\Omega_{\min}, \Omega_{\max}\right] \qquad \Omega_1, \Omega_2, \ldots, \Omega_N$$

And are then used to compute the PSD values and sinusoid amplitudes for each component:

$$\Phi(\Omega_i) = \Phi_0 \left(\frac{\Omega_i}{\Omega_0}\right)^{-w} \qquad A_i = \sqrt{2\Phi(\Omega_i)\Delta\Omega}$$

The random phases are then generated to finally construct the spatial road profile:

$$z_R(s) = \sum_{i=1}^{N} A_i \sin(\Omega_i s - \psi_i).$$

Which is then converted to a time-domain road input using the travelled distance:

$$s(t) = \int_0^t v(\tau)\, d\tau, \qquad z_R(t) = z_R(s(t)).$$

In the simulation, $s(t)$ is obtained numerically from the speed profile using cumulative integration (`s_profile = cumtrapz(t, v_profile);`). The speed profile is assigned by hand, just like the road patches/segments, so that different driving scenarios for testing and validating the Estimator can be created as intended. This generated road signal is then used as the vertical disturbance input to the quarter-car suspension model.

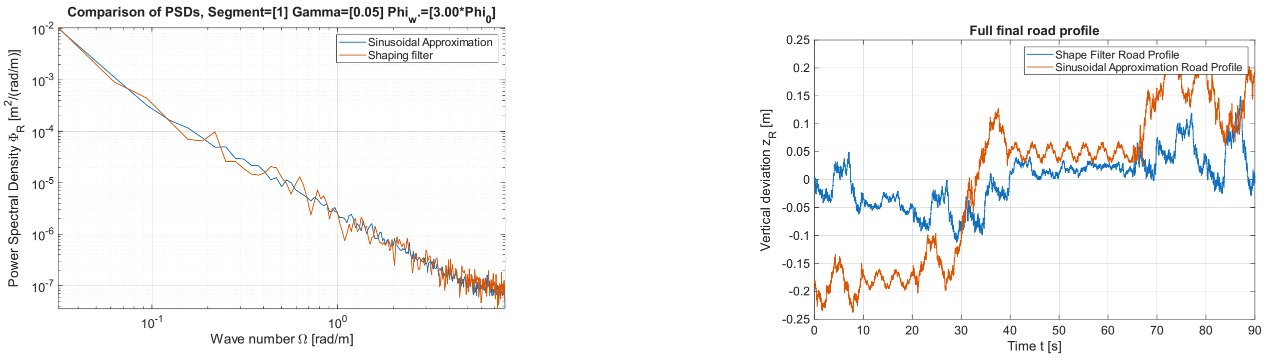
2.7.1 Alternative Road Profile using Shaping Filter

In the case of the Road Profile using a shaping filter (Sec. 2.6), the procedure is slightly different. The Road Profile z_R is generated by processing the road segment-by-segment. Each segment is associated with different road parameters, such as the main vehicle speed $v_{segment}$ and the roughness coefficient Φ_0. For every segment, the corresponding time interval is extracted from the global time vector t, and a local time vector starting from zero is created. The road elevation for the current segment is then generated using the shaping-filter model through the function `generateRoadProfileShapingFilter`. The spatial discretization step is computed as:

$$\Delta s = v_{segment}\, dt$$

assuming a constant vehicle speed and allows the stochastic differential equation to be simulated along the traveled segment.

To guarantee continuity between adjacent segments, the generated local profile is shifted so that its initial value matches the final value of the previous segment. Finally, all locally generated profiles are concatenated into the global vector `zR_shapefilter`, producing a continuous road profile over the entire simulation horizon. A shaping filter road profile compared to an original PSD road profile, using the same parameters for generation, are shown in Fig. 3.



(a) PSD comparison of the road profiles (of a segment).

(b) Full road profiles comparison.

Figure 3: Comparison of the two road profile generation methods.

3 Vehicle and Sensor Model

The vehicle is modeled using a quarter-car suspension model. This model includes the vertical motion of the sprung mass, representing the vehicle body, and the unsprung mass, representing the wheel assembly. The dynamical equations for this model are:

$$\boxed{\begin{aligned} m_s\ddot{z}_s &= -c_s(\dot{z}_s-\dot{z}_u)-k_s(z_s-z_u), \\ m_u\ddot{z}_u &= c_s(\dot{z}_s-\dot{z}_u)+k_s(z_s-z_u)-k_t(z_u-z_R) \end{aligned} \qquad \text{with state} \qquad \boldsymbol{x} = \begin{bmatrix} z_s & \dot{z}_s & z_u & \dot{z}_u \end{bmatrix}^\top .}$$

The estimator does not directly measure the road profile. Instead, it receives noisy measurements from the vehicle response. The proposed measurement vector is

$$\boldsymbol{y}_k = \begin{bmatrix} \ddot{z}_s(k) \\ z_s(k) - z_u(k) \end{bmatrix} + \boldsymbol{v}_k,$$

where $\ddot{z}_s$ is the chassis acceleration, $z_s - z_u$ is the suspension deflection, and $\boldsymbol{v}_k$ is measurement noise.

4 MCMC Estimation Problem

4.1 Bayesian Formulation

The goal is to estimate the road parameter vector $\boldsymbol{\theta} = [\Phi_0,\, w]^\top$ from noisy vehicle sensor data $y_{1:N}$. The problem is formulated using Bayes' theorem:

$$p(\boldsymbol{\theta} \mid y_{1:N}) \propto p(y_{1:N} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{1}$$

where the left-hand side is the *posterior* and the right-hand side is the product of the *likelihood* and the *prior*. Since the posterior cannot be computed directly, MCMC is used to draw samples from it without needing its normalizing constant.

4.2 Prior

A uniform prior is placed on $\boldsymbol{\theta}$ in the transformed space $\mathbf{q} = [\log_{10} \Phi_0,\, w]^\top$, within bounds covering all ISO road classes:

$$\log_{10} \Phi_0 \in [-6.4,\, -3.4], \qquad w \in [1.45,\, 2.55]. \tag{2}$$

Any proposal outside these bounds is immediately rejected. We decided to work in log-space for Φ_0 since the parameter spans several orders of magnitude across road classes.

4.3 Likelihood

For each estimation window, four features are extracted from the measured sensor signals: the RMS chassis acceleration $\sigma_{\ddot{z}_s}$, the RMS suspension deflection $\sigma_{z_s - z_u}$, and the RMS chassis acceleration in the $[0.5, 3]$ Hz and $[3, 15]$ Hz bands, denoted $\sigma_{\ddot{z}_s}^{\text{low}}$ and $\sigma_{\ddot{z}_s}^{\text{high}}$ respectively. All of them are log-transformed and collected into the feature vector:

$$\mathbf{f}_{\text{meas}} = \left[\log \sigma_{\ddot{z}_s},\ \log \sigma_{z_s - z_u},\ \log \sigma_{\ddot{z}_s}^{\text{low}},\ \log \sigma_{\ddot{z}_s}^{\text{high}}\right]^\top. \tag{3}$$

For any proposed $\boldsymbol{\theta}$, the corresponding predicted features are computed analytically. Since the quarter-car model is linear, the output variance under a given road PSD is:

$$\sigma_y^2 = \int_0^\infty |G(j\omega)|^2\, \Phi_R(\Omega)\, d\Omega, \qquad \omega = v\,\Omega, \tag{4}$$

where $G(j\omega)$ is the transfer function from road input to the sensor output of interest, evaluated from the quarter-car state-space model. This avoids generating any random road realization for each candidate $\boldsymbol{\theta}$, making the likelihood evaluation faster. Assuming independent Gaussian feature errors, the log-likelihood is:

$$\log p(y \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_{i=1}^{4} \left(\frac{f_{\text{meas},i} - f_{\text{pred},i}(\boldsymbol{\theta})}{\sigma_{f,i}} \right)^2. \tag{5}$$

4.4 Metropolis–Hastings Sampler

At each iteration, a new candidate $\mathbf{q}_{\text{prop}}$ is drawn by adding Gaussian noise to the current chain state:

$$\mathbf{q}_{\text{prop}} = \mathbf{q}_{\text{current}} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}\big(\mathbf{0},\, \operatorname{diag}([0.035,\, 0.025]^2)\big)\,. \tag{6}$$

The candidate is accepted with probability $\alpha = \min(1,\, e^{\Delta \log p})$, where $\Delta \log p$ is the log-posterior difference between the proposal and the current state. Better proposals are always accepted; worse ones are accepted with a probability that decreases with how much worse they are. This occasional acceptance of worse proposals allows the chain to explore the full posterior rather than getting stuck at a local maximum.

Each window runs a chain of 1200 iterations, discarding the first 300 to reduce the sensitivy from the initial condition. The posterior mean of the remaining samples gives the estimates $\hat{\Phi}_0$ and $\hat{w}$. Each new window initializes the chain from the previous window's estimate rather than a fixed starting point.

5 Suspension Mode Selection

After estimating the road parameters, the result $\hat{\boldsymbol{\theta}} = \begin{bmatrix} \hat{\Phi}_0 \\ \hat{w} \end{bmatrix}$ is mapped to a road class and then to a suspension mode.

Estimated road condition	**Interpretation**	**Suspension mode**
Low $\hat{\Phi}_0$	Smooth road / highway	Firm or cruise mode
Medium $\hat{\Phi}_0$	Normal road	Normal mode
High $\hat{\Phi}_0$	Rough or damaged road	Soft or raised mode

To avoid unnecessary switching of the drive settings, for example in the case of encounter of a short patch of road in different conditions, the mode is changed according to the rolling average of the road conditions encountered recently by the car, but this implementation is not demonstrated here due to the page constraint for the project. This rolling average controller, was however tested for a 90-second trial of intentionally alternated road class segments, to demonstrate this robustness capability, but is not visible in the 30-second trial results shown here.

6 Results and Performance Evaluation

The Estimator + Controller (Regulator) Schematic described in the sections above is illustrated in the diagram below. We now report and interpret the results of our simulation.

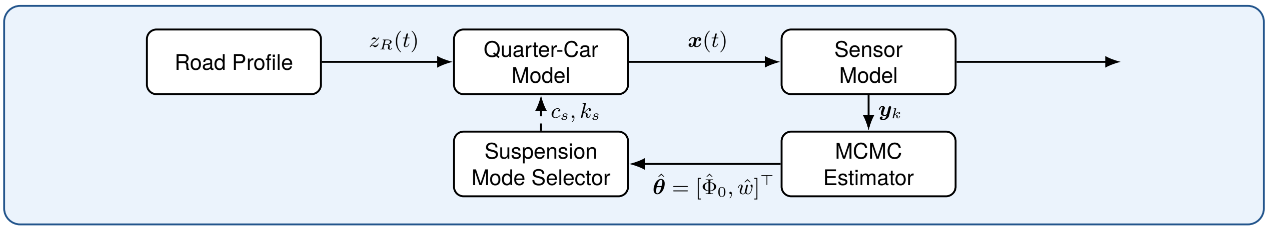


Figure 4: Proposed MCMC Estimator + Mode Selector Control Schematic

6.1 Simulated Scenario and Sensor Data

The simulation covers a 30-second drive across three road segments:

- urban: Class C, $\Phi_0 = 16 \times 10^{-6}$, $v = 45$ km/h
- highway: Class B, $\Phi_0 = 2 \times 10^{-6}$, $v = 80$ km/h
- rough rural: Class D, $\Phi_0 = 70 \times 10^{-6}$, $v = 20$ km/h

each segment with three superimposed potholes. The true Φ_0 spans nearly two decades, providing a demanding test for the estimator. Figure 5 shows the generated road profile, speed schedule, and true roughness parameter.

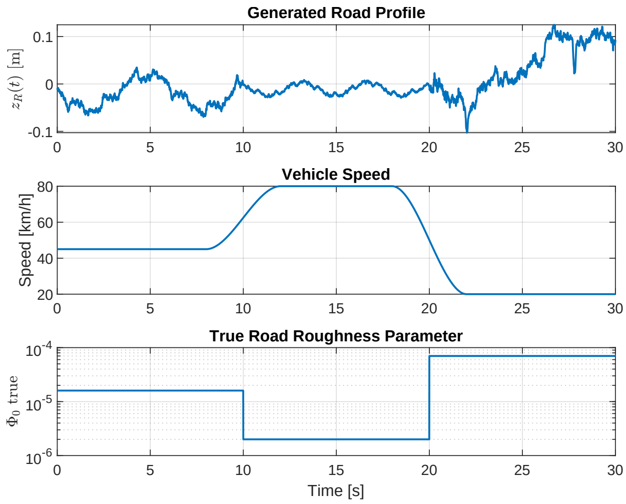


Figure 5: Generated road profile (top), speed profile (middle), and true Φ_0 (bottom).

6.2 Parameter Estimation

Figure 6 compares the true and estimated values of Φ_0 and w across all 14 estimation windows.

The roughness parameter Φ_0 is tracked well across all three segments. The two largest errors occur at transition windows, where the 4-second window straddles two different road classes, blending their statistics. The waviness w shows higher variability, as expected given its weaker influence on the feature vector. The overall validation metrics are reported in Table 2.

The mean log-roughness error of 0.107 decades corresponds to a factor of ≈ 1.28 on Φ_0. The two misclassified windows (at the D/C segment boundary and a late rough-road overshoot into Class E) are both correctly mapped to Comfort/Raised mode, which explains why road class accuracy is 85.7% while mode selection accuracy reaches 100%. The mode boundaries are intentionally coarser than the class boundaries precisely to absorb this kind of near-boundary estimation error.

Metric	Value
Mean $\lvert\Delta \log_{10}(\Phi_0)\rvert$	0.107 decades
Mean $\lvert\Delta w\rvert$	0.180
Road class accuracy	85.7 %
Mode selection accuracy	100.0 %
Mean MCMC acceptance rate	0.89

Table 2: Validation summary over all 14 estimation windows.

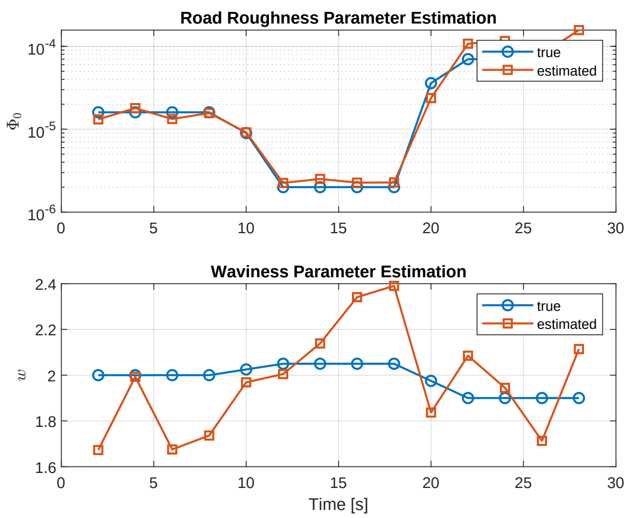


Figure 6: True versus estimated Φ_0 (top, log scale) and w (bottom) per window

6.3 MCMC Diagnostics and Adaptive Mode Selection

MCMC Diagnostics

The acceptance rate is consistently between 0.87 and 0.90 across all windows, confirming that the proposal step sizes are well matched to the posterior geometry. Figure 7 shows the posterior histograms for three representative windows.

The posteriors are unimodal in all cases. The highway window produces the sharpest posterior, as the smooth road gives a distinctive low-amplitude response that is easy to localise. The rough-road window is wider, consistent with the additional variability introduced by the potholes, which lie outside the stationary Gaussian model assumed by the estimator.

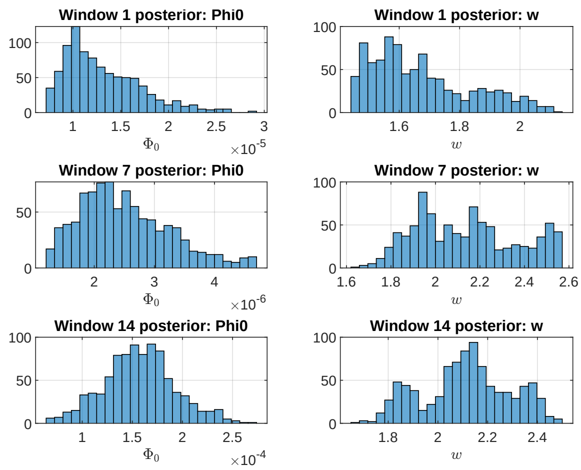


Figure 7: Post burn-in posterior histograms of Φ_0 and w for windows 1 (urban), 7 (highway), and 14 (rough rural).

Mode Selection and Adaptive Response

Figure 8 shows the selected suspension mode over time. The estimator correctly outputs Normal during the urban segment, switches to Firm/Cruise at $t = 12$ s, and to Comfort/Raised at $t = 22$ s. The approximate 2-second detection delay at each transition is a consequence of the sliding window.

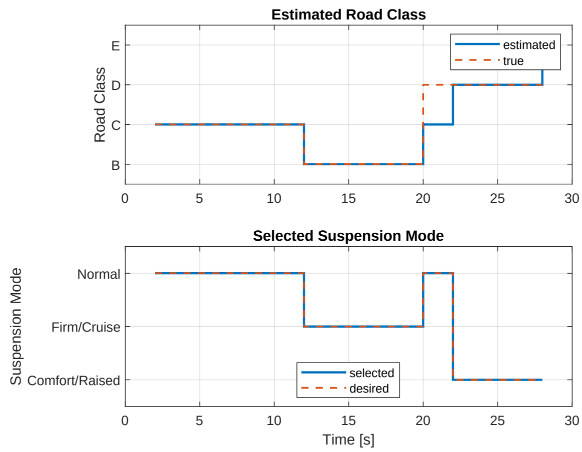


Figure 8: Estimated (solid) and true (dashed) road class and suspension mode.

Table 3 compares the global vehicle response under fixed-normal and adaptive damping. In the adaptive scenario, the suspension damping is switched in real time according to what the estimator decided for each window. The adaptive system reduces the RMS body acceleration by 5.3%, at the cost of a small increase in suspension deflection.

The segment-wise breakdown in Figure 9 shows that the acceleration improvement is concentrated in the highway segment and in the rural segment, while the urban segment sees little change as Normal mode is selected in both cases.

Metric	Fixed Normal	Adaptive
RMS body acceleration [m/s^2]	1.3342	1.2633
RMS suspension deflection [m]	0.01050	0.01172
Max suspension deflection [m]	0.05382	0.05934

Table 3: Global response comparison between fixed-normal and adaptive suspension.

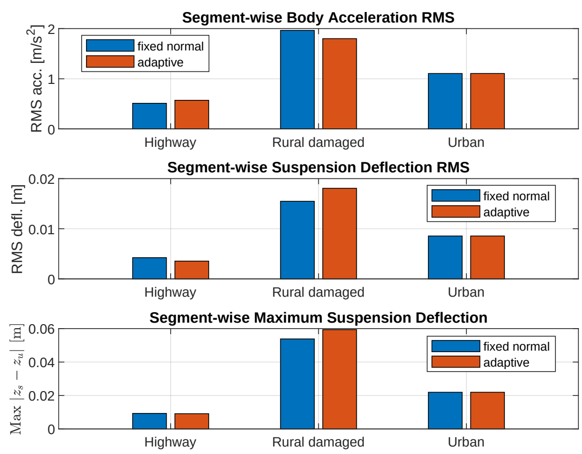


Figure 9: Segment-wise acceleration, RMS deflection, and maximum deflection for fixed-normal and adaptive configurations.

References

[1] G. Rill, *Vehicle Dynamics: Lecture Notes*, Hochschule Regensburg, University of Applied Sciences, March 2009.